

Phase 1 - Requirement Analysis and Planning

Project: Leave Management System (ServiceNow Scoped Application)

Status:  Completed

Owner: A.Chanikya

1. Project Overview

Before any table was created or any script was written, this project started from a single business problem statement: HR had no centralized, reliable system for employees to request leave and for the organization to track how much leave each employee actually had left. Leave data lived in a mix of spreadsheets, email threads, and manager memory which meant balances were frequently wrong, approvals were inconsistent across teams, and HR had no single source of truth to report from.

The Leave Management System was scoped to directly solve that: a ServiceNow scoped application that becomes the **single global system of record** for leave submission, approval, and balance tracking built so that the balance shown to an employee is always the actual, current, trustworthy number, not a snapshot someone updated manually last week.

2. Business Objective

HR Business has a requirement to create a new Application in ServiceNow which will be the global application for submitting leave requests by employees and maintaining leave data of employees.

This objective carries two distinct requirements that shaped the entire architecture: it must be **global** (usable across every office/country the company operates in, not built for one region and patched later) and it must **maintain leave data**, meaning balance tracking is a first-class function of the system, not an afterthought bolted onto a request form.

The organization operates across multiple geographies with different statutory leave entitlements, so a second, equally important requirement emerged early: the system must support **configuration of leave policy per country** — this

single requirement is the reason the Leave Calculator table exists as a distinct entity from the Leave Bucket, rather than just hardcoding leave entitlements into the application logic.

3. Business Requirements

ID	Requirement	Why It Matters
BR-01	System must allow employees to submit leave requests globally, from any location.	Core self-service capability — removes HR as a bottleneck for every single request.
BR-02	System must maintain a running, accurate leave balance per employee, per leave type.	Without this, the system is just a request log, not a management system — balance accuracy is the entire value proposition.
BR-03	System must allow HR/Admin to configure leave policy (number of days, validity period) per country.	Statutory leave entitlement varies by country; hardcoding this would require a code change for every new region.
BR-04	Leave requests must route to the employee's manager for approval before balance is deducted.	Prevents unauthorized leave from silently consuming balance; keeps the manager as the accountable decision-maker.
BR-05	System must prevent submission of leave requests exceeding available balance.	Stops the single most common data-integrity failure in manual leave tracking: negative or over-drawn balances.
BR-06	System must maintain an auditable snapshot of balance at the time of each request.	HR and managers need to be able to answer "what did this employee's balance look like when they applied," even after later requests have changed the live balance.

4. Stakeholders

Stakeholder	Role	Interest
HR Business Team	Requester / Process Owner	Accurate leave policy enforcement, clean data for reporting and compliance

Stakeholder	Role	Interest
Employees	End Users	A fast, self-service way to request leave and see their real balance at any time
Managers	Approvers	A clear, low-friction approval decision point with full context (dates, type, balance) at a glance
ServiceNow Admin/Developer	Builder	A maintainable, extensible configuration that doesn't require code changes for routine policy updates

Each stakeholder group's needs directly shaped a design decision downstream: the HR team's need for accurate reporting drove the audit-snapshot requirement (BR-06); the managers' need for low-friction decisions drove the choice to route approvals through native ServiceNow Approval records (visible in their existing Approvals list) rather than requiring them to open the Leave Request form directly.

5. Scope

In Scope

- Custom scoped application with a 3-table data model (Leave Request, Leave Bucket, Leave Calculator)
- Fully automated business logic for request numbering, duration calculation, balance validation, and balance deduction — no manual data entry into system-calculated fields
- Manager-based approval routing via Flow Designer, dynamically resolved per employee (no hardcoded approver)
- Country-wise leave policy configuration foundation via the Leave Calculator table
- Service Portal-facing request form, confirmation page, and role-based dashboards for employees and managers
- Role-based security model (Employee, Manager, HR Admin) enforced through ACLs, not just UI-level restrictions

Out of Scope (v1)

- Payroll system integration (leave data does not currently flow into a payroll calculation engine)
- Public holiday calendar exclusion from duration calculation — v1 uses calendar-day duration by explicit business decision, with working-day exclusion identified as a future enhancement
- Multi-level/hierarchical approval chains (v1 supports single-manager approval only)
- A dedicated native mobile app (the Service Portal is Now Mobile-compatible, but no separate mobile app was built)

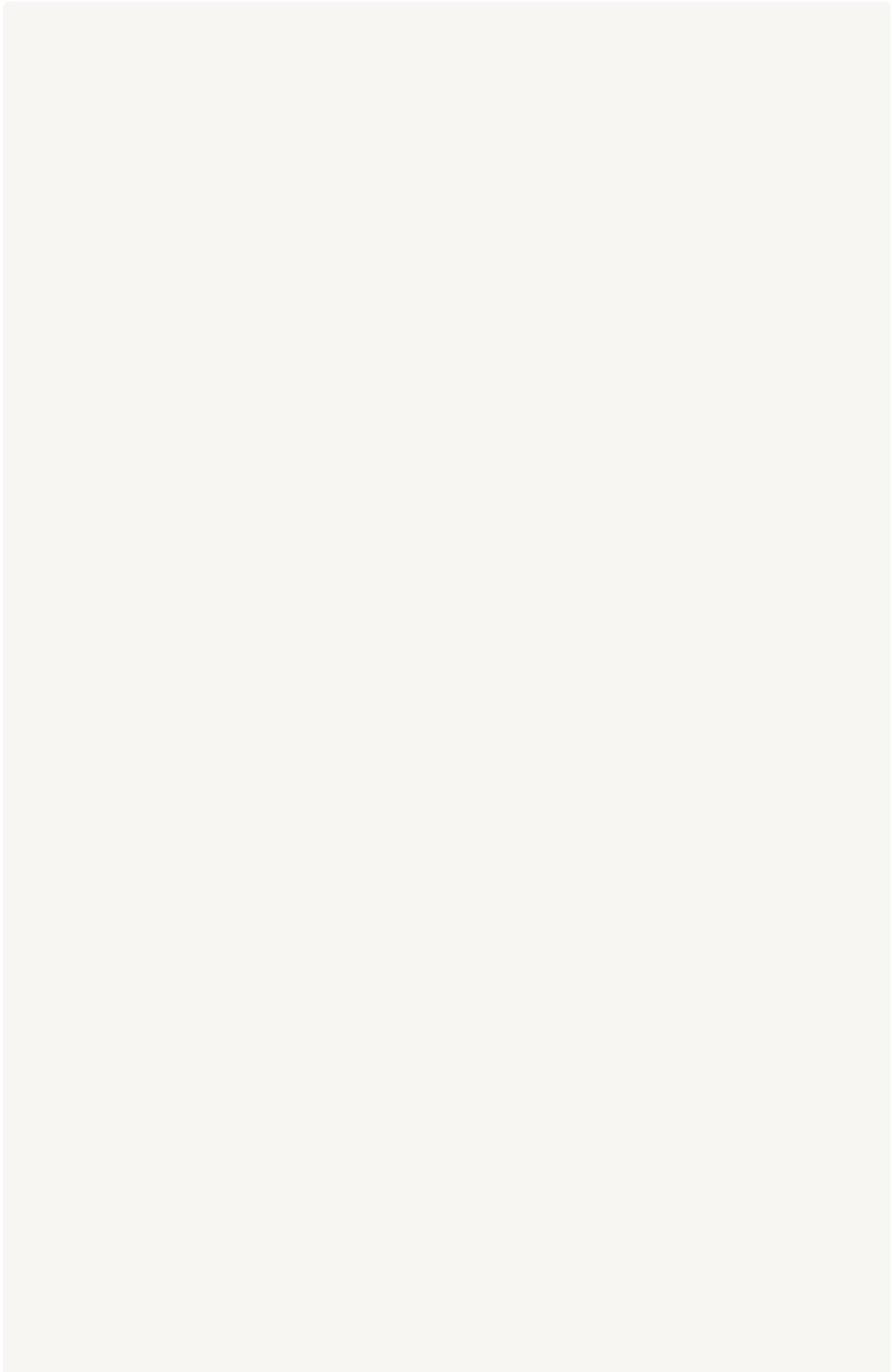
Scoping these items *out* explicitly, rather than leaving them ambiguous, was a deliberate decision — it gave the project a clear, achievable v1 boundary and created a documented, intentional backlog for future phases rather than scope creep discovered mid-build.

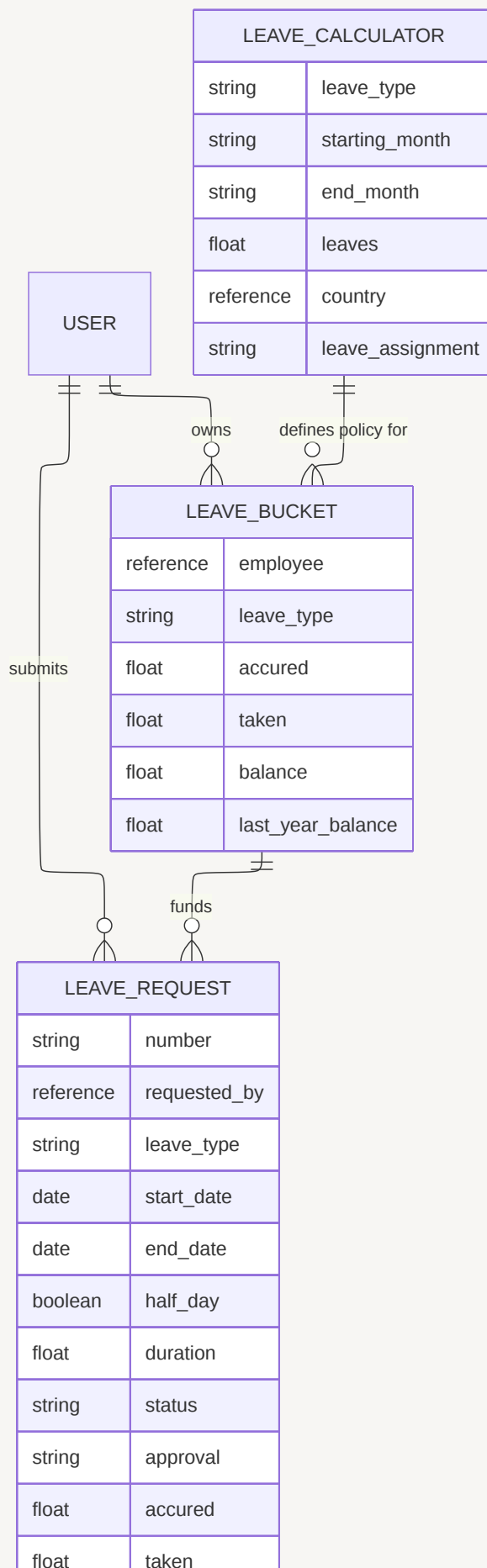
6. High-Level Data Model

The data model follows a three-tier pattern common in ledger-style systems: a **Policy** layer that defines the rules, a **Ledger** layer that holds the current state, and a **Transaction** layer that records individual events against that state.

Table	Tier	Purpose
Leave Calculator	Policy	Defines leave entitlement per Leave Type, per Country, per validity period — the configurable "rulebook" HR maintains
Leave Bucket	Ledger	One record per Employee + Leave Type; holds the live, current Accrued/Taken/Balance — the "wallet"
Leave Request	Transaction	Individual employee-submitted requests; holds a frozen audit snapshot of balance at the moment of submission

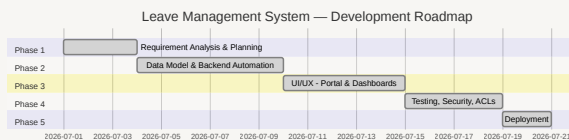
This separation is what allows the Leave Bucket to remain a fast, simple, always-current lookup, while the Leave Request table independently preserves historical accuracy — an employee's balance snapshot on a six-month-old request stays exactly as it was, even though their live balance has since changed many times over.





float	balance
string	comments

7. Project Roadmap



8. Success Criteria

Defining success criteria up front — before a single Business Rule was written — gave the project objective, testable checkpoints instead of a vague "it should work well" standard:

- An employee can submit a valid leave request in under 2 minutes via the Service Portal, with no HR involvement required.
- Balance validation prevents 100% of over-limit submissions at the point of entry — not caught later in a report, but blocked at submission time.
- Manager approval routes automatically with zero manual reassignment needed, regardless of which employee or which manager is involved.
- Leave Bucket balances remain accurate at all times — balance is only ever deducted on approval, never on mere submission, and never drifts out of sync with the sum of approved requests.
- Zero direct-field edits are possible on system-controlled fields (Status, Approval) through any UI path — enforced through both UI Policy (user experience) and ACL (actual security), not one or the other alone.